

DbISCDP: An Empirical Study on Distance-based Nearest Neighbor Approaches for Credit Default Prediction

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Abstract—Financial institutions usually use expert systems based on artificial intelligence to make credit lending a successful venture. However, the problem is made worse because these systems are typically not tailored to the lender's requirements and the instances are frequently quite diverse. Because of this, understanding the existing data is crucial for better insight. Since distance measures play a pivotal role in machine learning analytics, the focus has been given to the distance metrics utilized in the k-Nearest Neighbour (k-NN) algorithm. As the underlying structure of the data used to differ for datasets, focus has been given primarily to credit risk datasets. As no prior studies have been done on finding optimal distance metrics for credit risk datasets, the objective of the study is to find it. An appropriate distance metric for credit risk datasets can accurately determine the closest neighboring points which will help in decision making. For this, seven distance metric families with thirteen different distance measures have been considered in this scope. To get better insight, an expert system was proposed named Distance-based Intelligent System for Credit Default Prediction or DbISCDP utilizing 10-fold cross-validation for training and testing. In terms of average accuracy, Cosine and Hassanat distances provide better results for German and Australian benchmark datasets, respectively.

Index Terms—Credit Risk Assessment, Machine Learning, Distance Metric, K-Nearest Neighbour

I. INTRODUCTION

Financial risk analysis has grown over the past few decades due to the rise in financial transactions. Studies that examine the risk of customer credit transactions in commercial banks have gained rapid attention. Studies indicate that the majority of the bank's reported loss is attributed to customer credit risk, which is a rapidly emerging concern in the banking industry [1]. Consequently, banks must utilize an assessment technique to ascertain their clients' credit risk. Banks face a constant challenge in discovering innovative methods to do business that are both low-risk and highly efficient, while

still generating profit. "Credit Default Prediction (CDP)" is considered a systematic tool used by lenders to assess a borrower's ability to clear the loan. The goal of CDP is not just to find if a loan application should be approved, but at the same time to determine how much credit can be extended, how much time may be allowed, etc. CDP is vital for both classic and contemporary credit operations, such as client selection, risk measurement, loan monitoring, evaluation of overall credit performance, and portfolio risk management [2]. For these, various attempts have been made over a long period by financial organizations, credit managers, lenders, as well as academics, and financial economists, to develop an appropriate credit default prediction model [3]. The credit monitoring mechanism was defined as a 'True-False' binary problem, as per the previous studies, by classifying credit into two categories: credit-worthy and credit-unworthy [4]. The default risk was evaluated by the underwriters prior to the implementation of the automated credit monitoring system. Typically, a lending decision was reached through the judgment process, and the administrative authority evaluated applicants in accordance with the 5C's criteria[4], [5]. Subsequently, CDP [7] implemented several conventional statistical classification methodologies, including Logistic Regression (LR)[6], [7] and Discriminate Analysis (DA)[7]. Subsequently, due to the rigorous computational demands, artificial intelligence methodologies, including Support Vector Machine (SVM)[8], Neural Networks (NN)[2], [4], [5], Decision Trees (DT)[9], Random Forest (RF)[10], and Gradient Boosting (GB)[11], were utilized in the credit risk modeling process. Machine learning, a subfield of AI, is built on the premise that computers can automatically learn new tasks, recognize patterns in data, and choose appropriate actions with little to no oversight from humans. Various strategic approaches have

been employed over time. The most prevalent strategies are supervised and unsupervised learning. k-NN is an example of supervised learning. The performance of the k-NN algorithm completely relies on the choice of distance metric. Therefore, different metrics have been utilized to find the best metric in terms of accuracy and precision. Since, a distance metric is one of the important tools in machine learning; it has several applications including data balancing, outlier detection, and so many others. Therefore, the primary goal of this research is to find the appropriate distance metric for the credit risk assessment dataset. For better understanding, the article has been structured in the following way. Starting from Section II which discusses the literature survey, Section III provides a concise overview of the k-NN algorithm. Following that, section IV focused on the predominant usage of distance-based methods in the literature. Section V discussed the datasets and the performance metrics utilized in the study respectively. Methodology and Results have been discussed in sections VI and VII respectively. The study concluded in section VIII by demonstrating the practical implementation of a powerful machine learning technique, specifically k-Nearest Neighbours (k-NN), using two benchmark credit risk datasets.

II. LITERATURE SURVEY

The importance of k-NN in machine learning analytics is very much. Therefore several studies have been done on k-NN using credit risk datasets. [12] compared primitive classifiers for credit risk. As per the study, accuracy was found to be 72.20% and 89.10% for the German and Australian datasets respectively. Machine learning classifiers have been utilized by [13] for credit risk analysis. Euclidean distance has been employed as a distance metric while using the k-NN classifier. As per their study accuracy of 72.20% and 89.10% were recorded for German and Australian datasets respectively. [14] utilized the German dataset for credit risk analysis. k-NN was employed for the experimental study. Particle Swarm Optimization (PSO) was utilized to find the value of 'k'. As per the study, 68.50% accuracy was recorded for the base model with k-NN, and the accuracy of k-NN with PSO was recorded as high as 73.50%. In another study, [1] utilized k-NN as a baseline study for which 70.80% accuracy was recorded. [15] utilized German, Australian, and Taiwan credit data for their experimental study. As per the record, the baseline model achieved 75.2% and 85.7% accuracy for German and Australian datasets respectively. In another work [4] utilized k-NN for their experimental in which accuracy was recorded as 66% and 62.80% for German and Australian datasets respectively. Almost all the studies utilized default parameters for classification work. Experimental work on finding appropriate distance metrics for credit risk assessment has not been found in the literature.

III. BRIEF INTRODUCTION TO K-NEAREST NEIGHBOURS

This study employed the k-NN method, which is a type of lazy learning, as described by [16]. By nature, it is a non-parametric approach. The k-NN classifier relies on the

“Euclidean distance” to measure the similarity between a test sample and the designated training samples. Figure 1 illustrates two distinct categorical data points. The fundamental concept of the k-NN algorithm is to select the k closest neighbors from the training data when predicting a new location. Subsequently, the forecast for the new location may be determined by calculating the mean of the values from its k-nearest neighbors [16]. The algorithm for k-NN is the following.

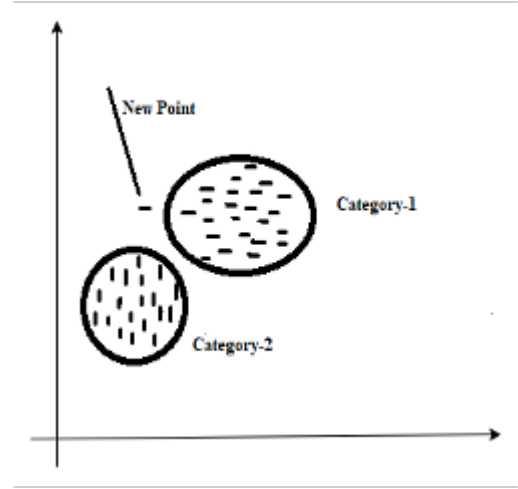


Fig. 1. Classification of a new point using k-NN.

Algorithm 1 k-Nearest Neighbors (k-NN) Classification

Input: Test data points, training data points, and a value of k
Output: Class labels for the test data points

- 1: **Step 1:** Choose the value of k
 - 2: **for** each test data point **do**
 - 3: **Step 2:** Calculate the distance between the test data point and each training data point
 - 4: **Step 3:** Sort the distance values in ascending order
 - 5: **Step 4:** Choose the top k values from the sorted list
 - 6: **Step 5:** Classify the test data point according to the majority class among these top k values
 - 7: **end for**
 - 8: **Step 6:** Repeat steps 2 to 5 for each test point
 - 9: **Step 7:** End
-

As we can observe the performance of the k-NN algorithm depends on the value of 'k' and the calculation of distance. As the datasets consist of 'n' dimensions, it's very difficult to say which distance metric will fit appropriately. Usually “Euclidian” distance is the first choice but it may not work properly for all cases. Therefore, a proper study is required to find the appropriate distance metrics as well as the 'k' value for credit risk datasets.

IV. DISTANCE MEASURING FAMILIES

A distance function estimates the distance between components of a set. If the distance between elements is zero,

they are equivalent; otherwise, they are distinct. The distance is a numerical representation that states how far apart things are. Distance and dissimilarity are synonyms for one another, as are proximity and nearness for similarity [16], [18]-[20]. An example of distances in two-dimensional space is shown in figure-2. The distance or dissimilarity function $d(x, y)$ between two vectors x and y determines the distance between both vectors as a non-negative real number. This function is considered a metric if it satisfies a list of conditions [16]-[18] that include the following:

- **Non-negativity:** $d_t(x, y) \geq 0$
- **Identity of indiscernibles:** $d_t(x, y) = 0 \iff x = y$
- **Symmetry:** $d_t(x, y) = d_t(y, x)$
- **Triangle inequality:** $d_t(x, y) \leq d_t(x, z) + d_t(z, y)$ where x, y , and z are vectors.

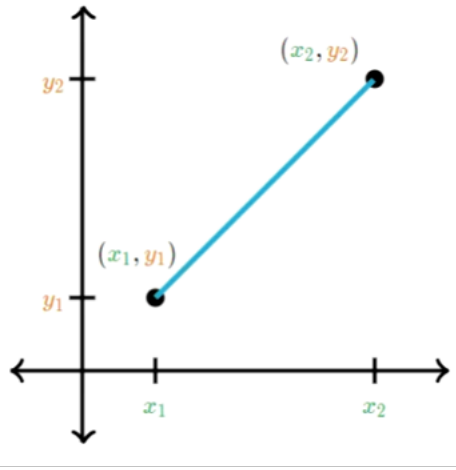


Fig. 2. An example of distances in 2D space.

Therefore, a distance function is simply a mathematical formulation that distance metrics employ. The distance function varies depending on the different distance metrics. The following distance metrics have been employed in this study to track changes in various credit risk datasets.

A. L_p Minkowski distance measures

This family of distances includes three distance metrics that are special cases of “Minkowski distance”, corresponding to different values of p . The “Minkowski distance”, which is also known as “ L_p norm”, is a generalized metric [16], [18], [20] given in equation 1.

$$d_t(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}} \quad (1)$$

where $x = (x_1, x_2, \dots, x_n)$ and $y = (y_1, y_2, \dots, y_n)$ are vectors, and p is a parameter that determines the type of distance metric. The distance definition can include the parameter p , which can take on different values. When $p = 2$, the distance becomes the Euclidean distance (L_2 Norm). “Euclidian distance” is the most well-known distance metric.

This metric relates to the straight line, which is the basic physical property of a distance between two points. The use of Euclidean distance in k-NN for data mining has the disadvantage of potentially being unsuitable for high-dimensional or multi-scale data. Euclidean distance is affected by noise and outliers and is susceptible to feature size and scale. When $p = 1$ it is known as Manhattan distance (L_1 Norm). The basic presumption is that the distance must be determined using the total of all absolute differences across all dimensions. The problem of “Manhattan distance” is that it may not correctly depict the similarities and differences between data points with different scales or dimensions. It is sensitive to the size and scope of the features. If two features have different ranges of values, the feature with the larger range will be dominant in the “Manhattan distance”. This might result in results that are biased or inaccurate when using k-NN. With $p=2$, “Minkowski” is Scikit-Learn’s default measure of distance, adopting Euclidian as the standard metric. “Euclidean” and “Manhattan” distances were represented in equations 2 and 3 respectively of Table-I.

B. L_1 distance measures

Many variations of the Manhattan distance metric have been proposed based on the L_1 norm. Based on the absolute difference, the family includes Lorentzian, Canberra, Sorensen, Soergel, Kulczynski, Mean Character, Non-Intersection distances [16], [18], [20]. To solve the normalization problem present in the Manhattan distance, the “Sorensen” distance, also known as the “BrayCurtis” distance, suggests normalizing the classical “Manhattan” distance by totaling the all components. Assuming that all vector values are non-negative, the Sorensen distance returns a value between 0 and 1, providing a more accurate interpretation of the distance value than the “Manhattan” distance. The Canberra distance which is also a modified Manhattan distance, is used to calculate the distance between two objects in a multi-dimensional space. It is defined as the total of the absolute differences between the two objects’ distances divided by the sum of the distances. It’s handy for comparing data that hasn’t been normalized or standardized. It is, however, sensitive to the sequence in which the objects are presented, which can have an impact on its performance in specific applications. This has the drawback of potentially having values that are almost zero for minute variations in the vector elements. Canberra and Sorensen distances were represented in equations 4 and 5 respectively of Table I.

C. Intersection Family

It includes metrics that are related to the intersection of X and Y . In this family, the $\min(X, Y)$ or $\max(X, Y)$ component can be found in either the denominator or the numerator part. Kulczynski distance also corresponds to the “Manhattan” distance with a normalization factor (divided by the $\max$ (or $\min$) of the coefficients). The “Motyka” distance is based on the maximum value instead of the difference, and the normalization is the sum of all coefficient pairs [16], [18],

TABLE I
DIFFERENT DISTANCE METRICS WITH EQUATIONS

Distance Family	Distance Metrics
A. L_p Minkowski	$EuclideanDistance : ED(x, y) = \sqrt{\sum_{i=1}^n x_i - y_i ^2} \quad (2)$ $ManhattanDistance : MD(x, y) = \sum_{i=1}^n x_i - y_i \quad (3)$
B. L_1 Distance Family	$CanberraDistance : CD(x, y) = \sum_{i=1}^n \frac{ x_i - y_i }{ x_i + y_i } \quad (4)$ $SorensenDistance : SD(x, y) = \frac{\sum_{i=1}^n x_i - y_i }{\sum_{i=1}^n (x_i + y_i)} \quad (5)$
C. Intersection Family	$MotykaDistance : MotD(x, y) = \frac{\sum_{i=1}^n \max(x_i, y_i)}{\sum_{i=1}^n (x_i + y_i)} \quad (6)$ $KulczynskiDistance : KD(x, y) = \frac{\sum_{i=1}^n x_i - y_i }{\sum_{i=1}^n \min(x_i, y_i)} \quad (7)$
D. Squared Chord Distance Family	$MatusitaDistance : MatD(x, y) = \left(\sqrt{\sum_{i=1}^n (\sqrt{x_i} - \sqrt{y_i})^2} \right)^2 \quad (8)$ $HellingerDistance : HelD(x, y) = \left(\sqrt{2 \sum_{i=1}^n (\sqrt{x_i} - \sqrt{y_i})^2} \right)^2 \quad (9)$
E. Inner-product Family	$JaccardDistance : JD(x, y) = \frac{\sum_{i=1}^n (x_i y_i)^2}{\sum_{i=1}^n x_i^2 + \sum_{i=1}^n y_i^2 - \sum_{i=1}^n x_i y_i} \quad (10)$ $CosineDistance : CSD(x, y) = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}} \quad (11)$
F. Squared L_2 Family	$Pearson(\chi^2)Family : PD(x, y) = \sum_{i=1}^n \frac{(x_i - y_i)^2}{y_i} \quad (12)$ $NeymanDistance : ND(x, y) = \sum_{i=1}^n \frac{(x_i - y_i)^2}{x_i} \quad (13)$
G. Others Distance Family	$HassanatDistance : HsntD(x, y) = \sum_{i=1}^n D(x_i, y_i) \quad (14)$ where $D(x, y) = \begin{cases} 1 - \frac{1 + \min(x_i, y_i)}{1 + \max(x_i, y_i)} & \text{if } \min(x_i, y_i) \geq 0 \\ 1 - \frac{1 + \min(x_i, y_i) + \min(x_i, y_i) }{1 + \max(x_i, y_i) + \min(x_i, y_i) } & \text{if } \min(x_i, y_i) < 0 \end{cases}$

[20]. Both the metrics are sensitive to the size and range of values of the features. “Motyka” and “Kulczynski” distances were represented in equations 6 and 7 respectively of Table-I.

D. Squared Chord Distance

“Squared Chord” distances are obtained by calculating the sum of the square root of the inner (dot) product which is referred to as the Fidelity similarity. The distances in this family cannot be used with features vector of negative values, this family includes “Bhattachayya”, “Squared Chord”, “Matusita”, “Hellinger” distances [16], [18], [20]. The “Matusita” and “Hellinger” distances are two illustrations of distance functions based on the Bhattacharyya coefficient, which evaluates the similarity between two probability distributions. One can utilize these distances to compare data points by treating each characteristic as a discrete probability distribution and comparing data points that have both numerical and category attributes. However, because they might not sufficiently take into consideration the relevant properties of similarity or dissimilarity between data points, these distances might not be optimal for all data sets and classification applications. “Matusita” and “Hellinger” distances were represented in equations 8 and 9 respectively of Table-I.

E. Inner-product Family

Distance measures from this family are determined by the product of pair-wise values from both vectors. This category encompasses the “Jaccard,” “Cosine,” “Dice,” and “Chord” distances [16], [18], [20]. The major disadvantage of the inner product is the lack of normalization. It is unclear to interpret it as small or large. As a result, different versions have been proposed, with the “cosine” similarity being the most prevalent, which may translate to a distance value between 0 and 1. “Jaccard” and “Cosine” distances were represented in equations 10 and 11 respectively of Table-I.

F. Squared L_2 Family

The L_2 distance measure family determines the total distance by considering the square of the difference at each point along both vectors. This family includes the following distance measures: Squared Euclidean, Clark, Neyman χ^2 , Pearson χ^2 , Squared χ^2 , Probabilistic Symmetric χ^2 , Divergence, Additive Symmetric χ^2 , Average, Mean Censored Euclidean, and Squared Chi-Squared distances [16], [18], [20]. Out of these, “Pearson” and “Neyman” distances were represented in equations 12 and 13 respectively of Table-I.

G. Other Distance Families

These metrics demonstrate distance methods that incorporate several concepts or measurements derived from prior distance measurements, these include and are not limited to “Average (L_1 , L_∞), Kumar-Johnson, Taneja, Pearson, Correlation, Squared Pearson, Hamming, Hausdorff, χ^2 statistic, Whittaker’s index of association, Meehl, Motyka and Hassanat distances” [16]-[20]. Out of these, “Hassanat” distance is represented in equation 14 of Table-I. The “Hassanat”

distance is confined inside the interval [0, 1]. The value of the function approaches 1 when the maximum value approaches infinity, assuming that the minimum value is finite, or when the minimum value approaches negative infinity, providing that the maximum value is finite. Equation 15 demonstrates this.

$$\lim_{\max(X_i, Y_i) \rightarrow \infty} D(X_i, Y_i) = \lim_{\max(X_i, Y_i) \rightarrow -\infty} D(X_i, Y_i) = 1 \quad (15)$$

The distances were established as a metric by fulfilling all of its metric properties. In this measure, regardless of the disparity between two numbers, the distance will always fall within the range of 0 to 1. As the dimension of the tested vectors rises, the maximum distance also increases, approaching the dimension itself. In the worst-case scenario, the distance increases linearly with the increase in dimensions.

V. DATASET DESCRIPTION AND PERFORMANCE METRICS

For this experimental study, two benchmark credit risk datasets named German¹ and Australian² Credit data have been utilized. The shapes of these datasets were 1000 x 21, and 690 x 16 respectively. In the German dataset, out of these 1000 instances, 700 were creditworthy instances and the rest 300 were the defaulters. For the Australian dataset, 36 instances contained NULL values. After removing the NULL values, 358 instances were good customers and 296 were defaulters.

The feature “class” represents the target class for binary classification, with a value of 1 for positive cases and 0 for negative classes. Two criteria, accuracy and precision, were employed to assess the performance of the basic model. Accuracy refers to the correct classification of classes given in equation 16. The variables FN, FP, TN, and TP represent the quantities of False Negative (FN), False Positive (FP), True Negative (TN), and True Positive (TP) respectively. Precision is the ratio of correctly reported positive identifications to all positive identifications, given in equation 17.

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP} \quad (16)$$

$$Precision = \frac{TP}{TP + FP} \quad (17)$$

VI. METHODOLOGY

A distance-based intelligent system for credit default prediction or DbISCDP has been introduced in this section. To find the appropriate distance metric, two benchmark credit data were collected from the UCI ML repository. Subsequently, datasets need to undergo pre-processing involving the removal of missing data, encoding, normalization, and other techniques such as feature selection and outlier detection to enhance interpretation. The dataset encoding and normalization were performed using the ‘labelencoding’ and ‘Minmaxscalar’ methodologies respectively. An iterative training and validation of the model using the dataset requires the execution of a train test

¹<https://doi.org/10.24432/C5QG88>.

²<https://doi.org/10.24432/C5FS30>.

split. The choice of a suitable machine-learning model depends on the nature of the research challenge. Upon completion of training and testing, it is necessary to apply performance metrics to evaluate the level of performance. Credit risk analysis has been addressed using the k-nearest neighbor algorithm in the experimental scenario. A train test split of 60:40 has been taken into account. Validation is being conducted using 10-fold cross-validation to enhance understanding. Finally, both the accuracy and precision scores have been documented. The structure of our model is illustrated in Figure 3. The k-NN algorithm evaluates several distance measurements as input and generates the optimal output by selecting the appropriate 'k' number through the elbow approach. Finally, the accuracy and precision were recorded using the best 'k' neighbors.

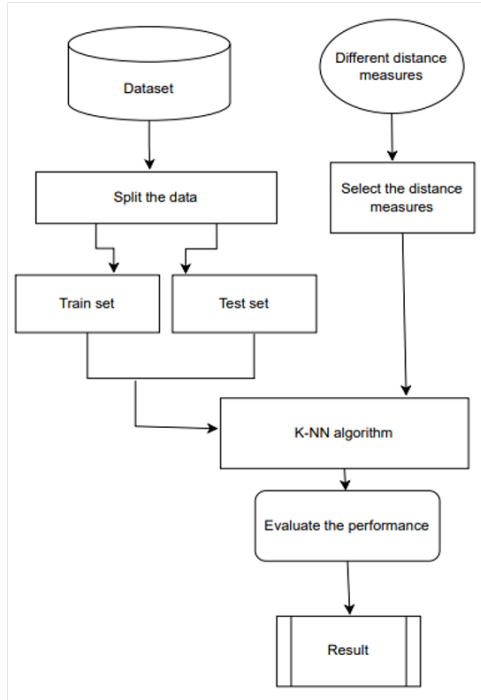


Fig. 3. Flowchart of DbISCDP

VII. RESULTS AND DISCUSSION

The experimental study was conducted utilizing Python 3.11.5 on an Apple M2-based system. The findings revealed huge variations between the performances of various distance measures, indicating that the KNN classifier's performance depended considerably on the distance utilized. For the German dataset, Sorenson achieved the lowest error and highest precision with 23 as 'k' in the k-nearest neighbor technique. A 10-fold CV was maintained to find the minimum error of 0.263, whereas maximum accuracy and precision were calculated utilizing a 60:40 train test split. Although the cosine distance metric achieved the highest accuracy (74.25%), Sorensen wasn't too far behind with 73.25%. From the results, it was evident that all the results were found for k in the range of 16 to 29 and the best performance was recorded at

23. For the Australian dataset, Canberra achieved the lowest error, highest accuracy, and precision as well with 8 as k in the k-nearest neighbor technique. The results were concentrated majorly for k between 4 to 16. Table II and III shows the results obtained from German and Australian Credit Risk datasets respectively.

TABLE II
PERFORMANCE OF DIFFERENT TYPES OF DISTANCES IN K-NN UTILIZING GERMAN CREDIT DATA

Distance Metrics	k-all	Minimum Error	Max Accuracy	Precision
1. Euclidean Distance	29	0.266	0.7075	0.6538
2. Manhattan Distance	23	0.265	0.7150	0.6571
3. Canberra Distance	16	0.2679	0.7300	0.6349
4. Sorensen /Bray Curti's Distance	23	0.263	0.7325	0.68
5. Motyka Distance	29	0.363	0.6675	0
6. Kulczynki's Distance	26	0.313	0.6700	0.1111
7. Hellinger Distance	23	0.3060	0.6750	0.2727
8. Matusita Distance	23	0.3060	0.6750	0.2727
9. Jaccard Distance	18	0.2679	0.7175	0.6071
10. Cosine Distance	17	0.269	0.7425	0.6718
11. Pearson χ^2 Distance	29	0.422	0.5750	0.2897
12. Neyman χ^2 Distance	29	0.304	0.6800	0.2
13. Hassanat Distance	24	0.274	0.7050	0.5466

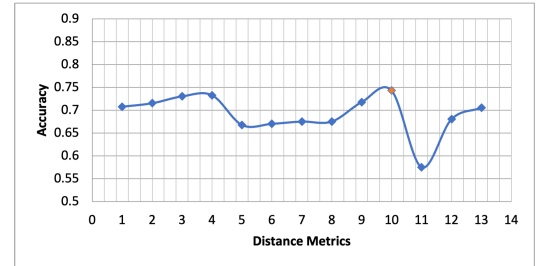


Fig. 4. Graphical representation of accuracy for all distance metrics utilizing German credit dataset

TABLE III
PERFORMANCE OF DIFFERENT TYPES OF DISTANCES IN K-NN UTILIZING AUSTRALIAN CREDIT DATA

Distance Metrics	k-all	Minimum Error	Max Accuracy	Precision
1. Euclidean Distance	16	0.1434	0.9618	0.9722
2. Manhattan Distance	24	0.1392	0.9618	0.9722
3. Canberra Distance	8	0.139	0.9733	0.9906
4. Sorensen /Bray Curti's Distance	18	0.1464	0.9618	0.9722
5. Motyka Distance	1	0.5159	0.2557	0.2601
6. Kulczynki's Distance	1	0.5152	0.1870	0.2356
7. Hellinger Distance	5	0.1420	0.9695	0.9814
8. Matusita Distance	5	0.1420	0.9695	0.9814
9. Jaccard Distance	1	0.5200	0.2519	0.2586
10. Cosine Distance	15	0.1481	0.9695	0.9814
11. Pearson χ^2 Distance	1	0.5152	0.1870	0.2350
12. Neyman χ^2 Distance	1	0.4724	0.1870	0.2356
13. Hassanat Distance	4	0.1405	0.9733	0.9906

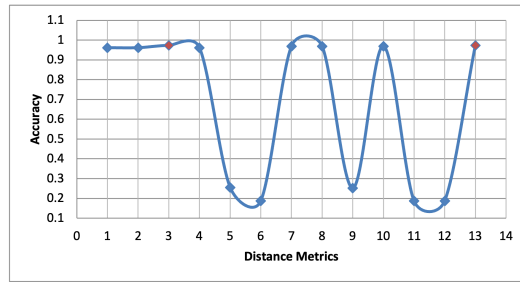


Fig. 5. Graphical representation of accuracy for all distance metrics utilizing Australian credit dataset

Figure-4 and 5, show the graphical representation of changes in accuracy recorded for German and Australian datasets respectively. The highest accuracy achieved metrics were represented in red dots. In majority of the existing machine learning models employed default parameters applicable for k-NN. With $p=2$, the metric is 'Minkowski', and the number of nearest neighbors is set to '5', as the default value implemented in Scikit-Learn results as 5 nearest neighbors utilizing the Euclidean distance measure. A comparison of existing nearest-neighbor models with the proposed method is shown in Table IV.

TABLE IV
COMPARISON OF EXISTING K-NN BASED MODELS UTILIZING GERMAN AND AUSTRALIAN CREDIT DATA

German Dataset		Australian Dataset	
Reference	Accuracy	Reference	Accuracy
[28]	72.20	[28]	89.10
[29]	72.20	[29]	89.10
[30]	68.50	[31]	86.75
[1]	70.80	[5]	62.80
[47]	72.66	Proposed Method	97.33
[5]	66.00		
Proposed Method	74.25		

VIII. CONCLUSION

Concerning distance measures and data sets, the performance evaluation of the kNN classification of credit risk data using various distance measures revealed significant variances. Among the distance families, L_p Distance Family, L_1 Distance Family, Inner-product Family, and Other Distance Families achieved constant performance for the datasets. Specifically, Hassanat, Canberra, Sorensen, and Cosine distance measures were among the group of distance measures that exhibited stable performance across the data sets. The results indicate that the value of 'k' in kNN, the dataset, and the distance metric significantly influenced the performance of the expert system. The research demonstrates that there is no singular optimal value for "k" or the distance measure across all data sets. This is why different distance measures and the number of nearest neighbors should be tested on the data sets before being chosen for a study. Although 10-fold cross-validation was employed in this study, 5-fold cv can also be tested. Various performance metrics beyond accuracy and precision can be employed for enhanced validation. Data balancing is another

scope of future work. The study employed thirteen distinct distance measures to enhance the classification performance of the k-NN algorithm. The analysis ultimately revealed the most effective distance measures for the datasets.

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